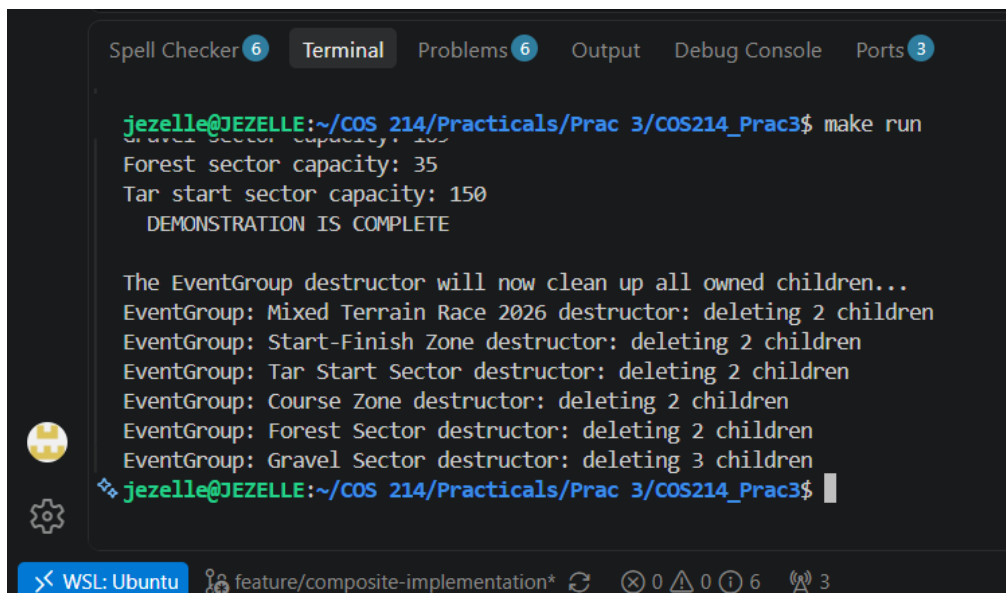




```
jezelle@JEZELLE:~/COS 214/Practicals/Prac 3/COS214_Prac3$ make run
Gravel Sector capacity: 100
Forest sector capacity: 35
Tar start sector capacity: 150
DEMONSTRATION IS COMPLETE

The EventGroup destructor will now clean up all owned children...
EventGroup: Mixed Terrain Race 2026 destructor: deleting 2 children
EventGroup: Start-Finish Zone destructor: deleting 2 children
EventGroup: Tar Start Sector destructor: deleting 2 children
EventGroup: Course Zone destructor: deleting 2 children
EventGroup: Forest Sector destructor: deleting 2 children
EventGroup: Gravel Sector destructor: deleting 3 children
jezelle@JEZELLE:~/COS 214/Practicals/Prac 3/COS214_Prac3$
```

Total objects deleted: 14 (1 RaceEvent + 2 RaceZones + 3 CourseSectors + 8 Leaves) which matches exactly with the objects created.

1. **Single Ownership:** Each component has exactly one owner parent (a single addChild() call was made for each object).
2. **Only the owner deletes** – In the recursive destructor from EventGroup, only children vector are deleted. Objects in the observers vector are not deleted, since it's non-owning pointers.
3. **Transfer Safety:** releaseChild() removes from old parent, addChild() adds to new parent, so the ownership transfers cleanly.
4. **Zero Double-Deletion:** An object is never in two children vectors simultaneously, so it's never deleted twice.
5. **Zero Memory Leaks:** Every object created with new is deleted exactly once by its owning composite (or by EventControl for the root).